Advanced Formal Tools

CTL Model Checker

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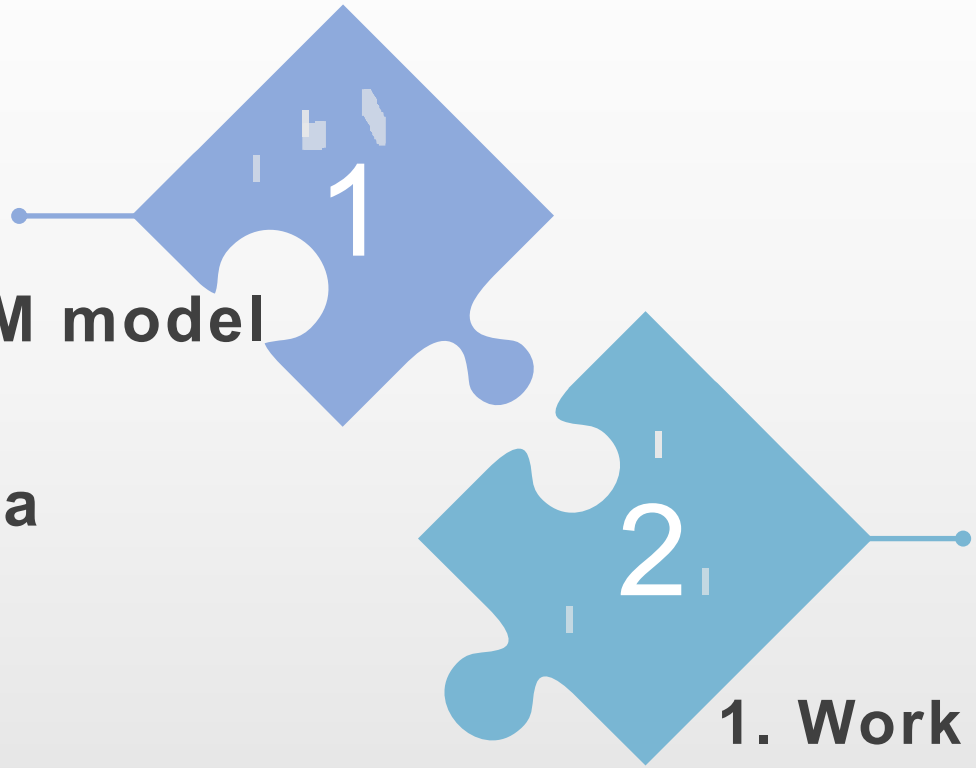
CONTENTS

Current Status

1.implement the FSM model reader

2.implement Formula Evaluation

3.Finish the testing

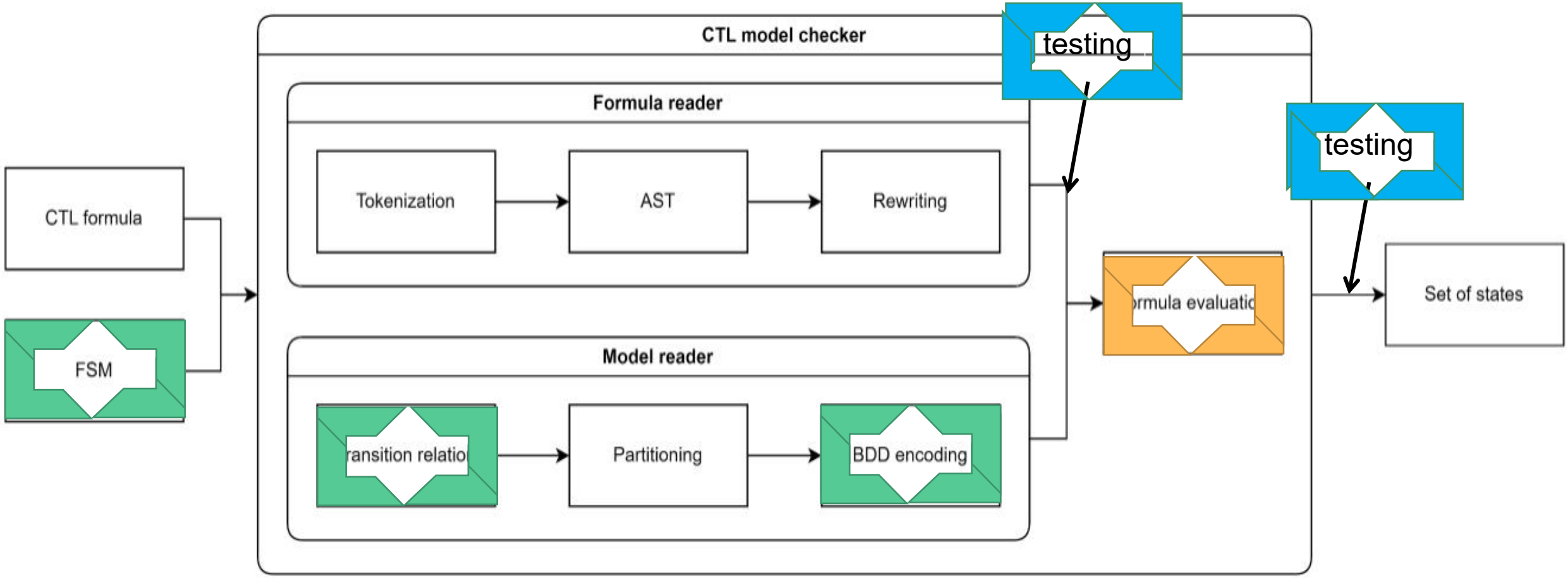


Next Goal

1. Work on Partitioning
2. Research different CTL models and compare

CTL model checking algorithm

--based on forward state traversal



We did -- FSM model reader

- **Specification of the FSM input format:**

  Number of states n, the first state is the initial state

  Next n lines: state, number of labels, list of labels

  Number of edges m

  Next n lines: state, state

- **Example of FSM input format:**

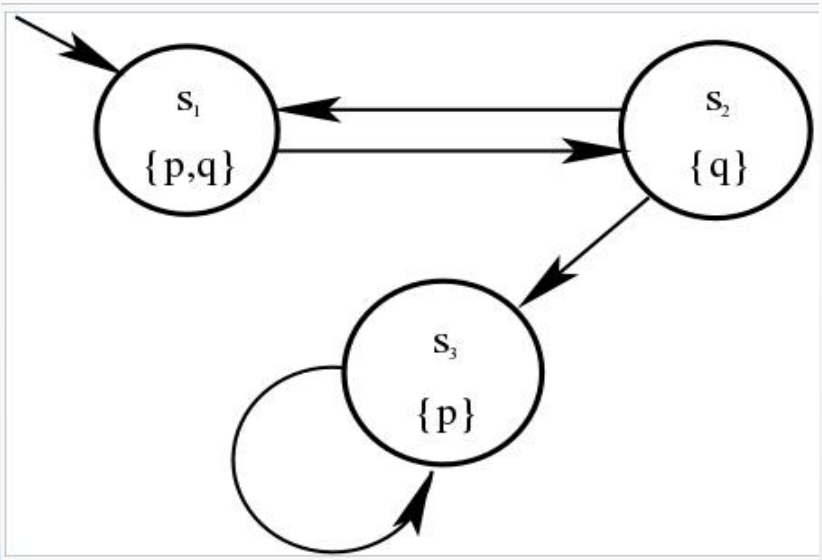
  3
  S0 2 begin start
  S1 0
  S2 1 finish

  2
  S0 S1
  S1 S2

We did -- FSM model reader

Example:

- 1. Kripke structure (model checking)



- 2. Kripke structure(transition system)

  M = (S, I, R, L)

  S = {s1, s2, s3}.
  I = {s1}.
  R = {(s1, s2), (s2, s1) (s2, s3), (s3, s3)}.
  L = {(s1, {p, q}), (s2, {q}), (s3, {p})}.

- 3. FSM input format:

  3
  S1 2 p q
  S2 1 q
  S3 1 p
  4
  S1 S2
  S2 S1
  S2 S3
  S3 S3

- 4. Transition relation

  R = {(s1, s2), (s2, s1) (s2, s3), (s3, s3)}.

  L = {(s1, p),(s1, q), (s2, q), (s3, p)}.

We did -- FSM model reader

Example:

- 3. Transition relation

R = {(s1, s2), (s2, s1), (s2, s3), (s3, s3)}
L = {(s1, p), (s1, q), (s2, q), (s3, p)}

- 4. Binary encoding

State	Binary Encoding
s1	00
s2	01
s3	10

State	Binary Encoding
p	00
q	01

- 5. Expressed as a Boolean function

state	s1’	s2’	s3’
00	0	1	0
01	1	0	1
10	0	0	1

state	s1’	s2’	s3’
00	1	0	1
01	1	1	0

We did -- Formula Evaluation

- Recursively traverse the AST
- Part1: True False Atom Not And p0 Eq Img
- Part2: EX EG EU FU FG

$$\mathbf{EX}f = Pref$$
$$\mathbf{E}[g\mathbf{U}f] = \mathbf{lfp}Z[f \vee (g \wedge \mathbf{EX}Z)]$$
$$\mathbf{EG}f = \mathbf{gfp}Z[f \wedge \mathbf{EX}Z]$$

$$FwdUntil(p, q) = \mathbf{lfp}Z\,[p \vee Img(Z \wedge q)]$$

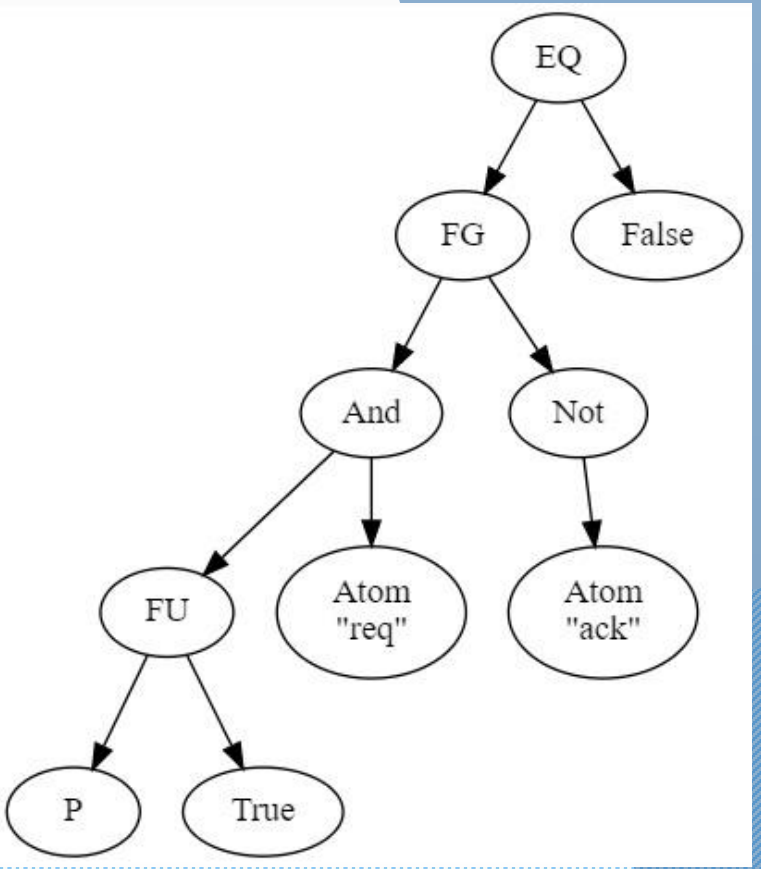
$$EH(p) = \mathbf{gfp}Z[p \wedge Img(Z)]$$
$$Reachable(p, q) = FwdUntil(p; q) \wedge q$$
$$FwdGlobal(p, q) = EH(Reachable(p, q))$$

We did -- Testing 1

- More than 20 tests to evaluate various CTL formulas against several different models
- Found several bugs thanks to tests

Example: AST ast("AG(req ⇒ AFack)");

$$
\begin{aligned}
& s_0 \models \mathbf{AG}\,(req \rightarrow \mathbf{AF}\,ack) \\
\iff\ & s_0 \models \neg\mathbf{E}\,[\mathrm{true}\,\mathbf{U}\,(req \wedge \mathbf{EG}\,\neg ack)] \\
\iff\ & p_0 \wedge \mathbf{E}\,[\mathrm{true}\,\mathbf{U}\,(req \wedge \mathbf{EG}\,\neg ack)] = \mathrm{false} \\
\iff\ & \mathit{FwdUntil}\,(p_0, \mathrm{true}) \wedge (req \wedge \mathbf{EG}\,\neg ack) = \mathrm{false} \\
\iff\ & (\mathit{FwdUntil}\,(p_0, \mathrm{true}) \wedge req) \wedge \mathbf{EG}\,\neg ack = \mathrm{false} \\
\iff\ & \mathit{FwdGlobal}\,((\mathit{FwdUntil}\,(p_0, \mathrm{true}) \wedge req), \neg ack) = \mathrm{false}.
\end{aligned}
$$



We did -- Testing 2

- More than 20 tests to evaluate various CTL formulas against several different models
- Found several bugs thanks to tests

Example: AG(EF(a)) and FSM model1

1. AST construction: To construct a syntax tree
AST ast("AG(EF(a))");

2. FSM model reader: To read a given state transition model and represent it as a state transition graph.
FSM fsm(model1);

3. Formula evaluation: Traverse the state transition graph, compute all state sets that satisfy the given CTL formula, and return the results.
EXPECT_TRUE(evaluate_formula(ast, fsm));

```
Model1:
4
S0 2 a b
S1 1 b
S2 1 b
S3 1 c
7
S0 S1
S0 S2
S1 S1
S1 S2
S2 S0
S2 S3
S3 S1
```

What we want to do next step?

- **FIRST:**

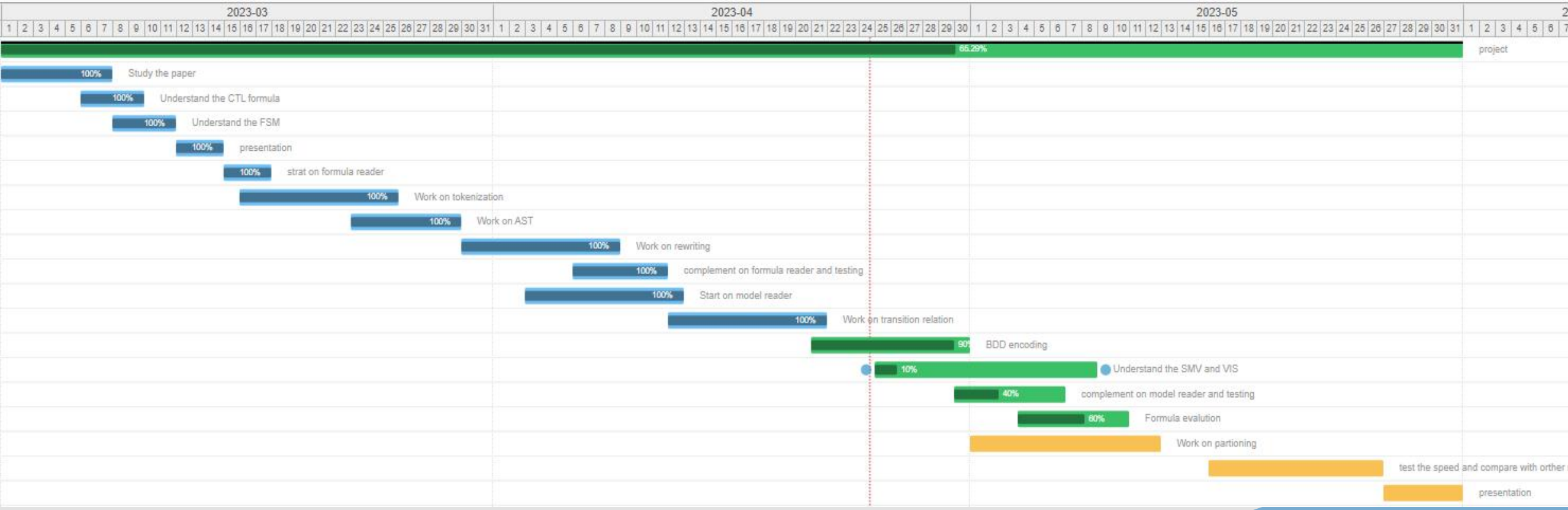
Implement partitioning

- **SECOND:**

Various performance comparisons

Study different CTL models and compare them

Schedule


2023-03
2023-04
2023-05
project
Study the paper
Understand the CTL formula
Understand the FSM
presentation
strat on formula reader
Work on tokenization
Work on AST
Work on rewriting
complement on formula reader and testing
Start on model reader
Work on transition relation
BDD encoding
Understand the SMV and VIS
complement on model reader and testing
Formula evalution
Work on partioning
test the speed and compare with orther
presentation

THANK YOU